3.3.1 Manufacture of chlorine:

Chlorine is manufactured by the electrolysis of brine in electrolytic process or by oxidation of HCl by air in Deacon's process.

Electrolytic process: When a solution of brine (NaCl) is electrolysed, Na^+ and Cl^- ions are formed. Na^+ ion reacts with OH^- ions of water and forms sodium hydroxide. Hydrogen and chlorine are liberated as gases.

$NaCl \longrightarrow Na^+ + Cl^-$

$H_2O \longrightarrow H^+ + OH^-$

$Na^+ + OH^- \longrightarrow NaOH$

At the cathode,

$H^+ + e^- \longrightarrow H$

$H + H \longrightarrow H_2$

At the anode,

$Cl^- \longrightarrow Cl + e^-$

$Cl + Cl \longrightarrow Cl_2$

Deacon's process: In this process a mixture of air and hydrochloric acid is passed up a chamber containing a number of shelves, pumice stones soaked in cuprous chloride are placed. Hot gases at about 723 K are passed through a jacket that surrounds the chamber.

$$4HCl + O_2 \xrightarrow[Cu_2Cl_2]{400^0C} 2H_2O + 2Cl_2 \uparrow$$

The chlorine obtained by this method is dilute and is employed for the manufacture of bleaching powder. The catalysed reaction is given below,

$$2Cu_2Cl_2 + O_2 \longrightarrow \underset{\text{Cuprous oxy chloride}}{2Cu_2OCl_2}$$

$$Cu_2OCl_2 + 2HCl \longrightarrow \underset{\text{Cupric chloride}}{2CuCl_2} + H_2O$$

$$2CuCl_2 \longrightarrow \underset{\text{Cuprous chloride}}{Cu_2Cl_2} + Cl_2$$

Physical properties:

Chlorine is a greenish yellow gas with a pungent irritating odour. It produces headache when inhaled even in small quantities whereas inhalation of large quantities could be fatal. It is 2.5 times heavier than air.

Chlorine is soluble in water and its solution is referred as chlorine water. It deposits greenish yellow crystals of chlorine hydrate ($Cl_2.8H_2O$). It can be converted into liquid (Boiling point – 34.6° C) and yellow crystalline solid (Melting point -102° C)

Chemical properties:

Action with metals and non-metals: It reacts with metals and non metals to give the corresponding chlorides.

$$2Na + Cl_2 \longrightarrow 2NaCl$$

$$2Fe + 3Cl_2 \longrightarrow 2FeCl_3$$

$$2Al + 3Cl_2 \longrightarrow 2AlCl_3$$

$$Cu + Cl_2 \longrightarrow CuCl_2$$

$$H_2 + Cl_2 \longrightarrow 2HCl \quad ; \quad \Delta H = -44kCal$$

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